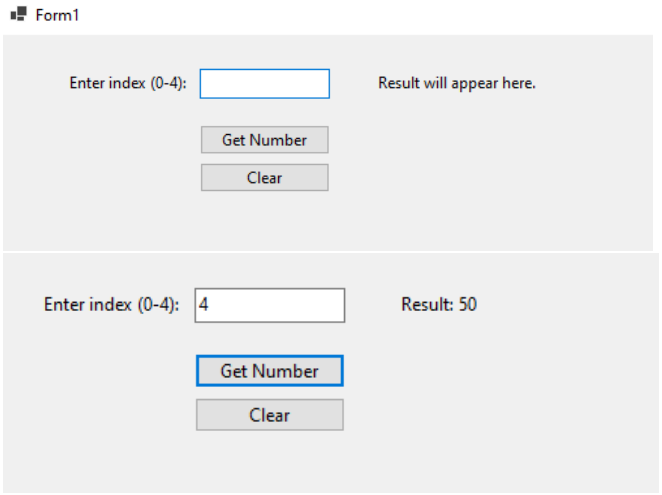
1. 

CODE: namespace WinFormsApp2

```
{
    public partial class Form1 : Form
    {
        public Form1()
        {
            InitializeComponent();
        }

        private void button1_Click(object sender, EventArgs e)
        {
            int[] numbers = { 10, 20, 30, 40, 50 };

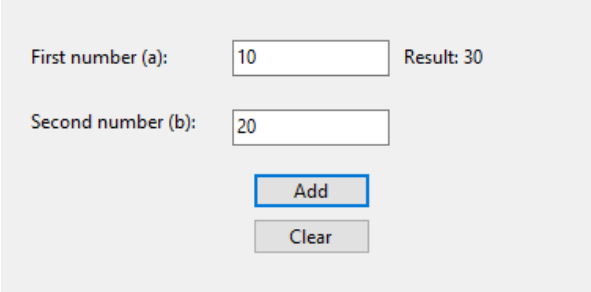
            int index;

            if (int.TryParse(txtInput.Text, out index))
            {
                if (index >= 0 && index < numbers.Length)
                {
                    int answer = numbers[index];
```

```
            lblResult.Text = "Result: " + answer;
        }
        else
        {
            lblResult.Text = "Please enter an index from 0 to 4.";
        }
    }
    else
    {
        lblResult.Text = "Please enter a valid number.";
    }
}

private void btnClear_Click(object sender, EventArgs e)
{
    txtInput.Clear();
    lblResult.Text = "Result will appear here.";
    txtInput.Focus();
}
}
}
```

2.



CODE:

```
namespace WinFormsApp2
{
```

```
public partial class Form1 : Form
{
    public Form1()
    {
        InitializeComponent();
    }

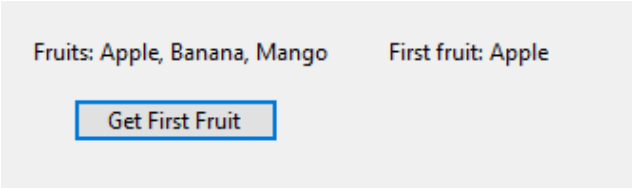
    private void button1_Click(object sender, EventArgs e)
    {
        int a;
        int b;

        if (int.TryParse(txtFirstNumber.Text, out a) &&
            int.TryParse(txtSecondNumber.Text, out b))
        {
            int sum = a + b;

            lblResult.Text = "Result: " + sum;
        }
        else
        {
            lblResult.Text = "Please enter valid whole numbers.";
        }
    }

    private void btnClear_Click(object sender, EventArgs e)
    {
        txtFirstNumber.Clear();
        lblResult.Text = "Result will appear here.";
        txtFirstNumber.Focus();
    }
```

```
    }
}
```



3.

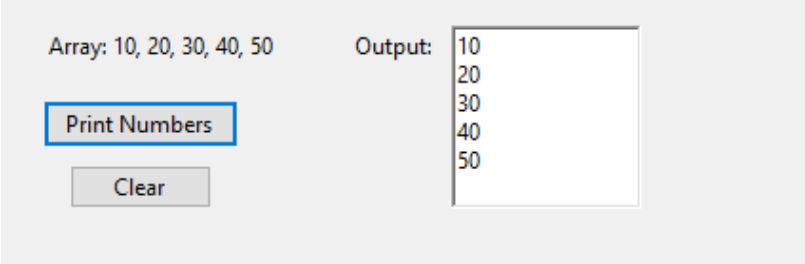
CODE:

```
namespace WinFormsApp2
{
    public partial class Form1 : Form
    {
        public Form1()
        {
            InitializeComponent();
        }

        private void button1_Click(object sender, EventArgs e)
        {
            string[] fruits = { "Apple", "Banana", "Mango" };

            string fruit = fruits[0];

            lblResult.Text = "First fruit: " + fruit;
        }
    }
}
```

4. 

CODE:

```
namespace WinFormsApp2
{
   public partial class Form1 : Form
   {
      public Form1()
      {
         InitializeComponent();
      }

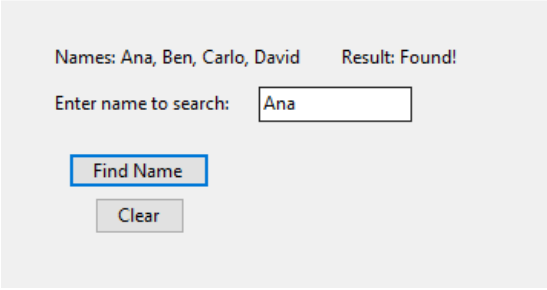
      private void button1_Click(object sender, EventArgs e)
      {
         int[] numbers = { 10, 20, 30, 40, 50 };

         rtbOutput.Clear();

         for (int i = 0; i < numbers.Length; i++)
         {
            rtbOutput.AppendText(numbers[i] + Environment.NewLine);
         }

      }

      private void btnClear_Click(object sender, EventArgs e)
      {
         rtbOutput.Clear();
      }
   }
}
```

5. 

CODE:

```
namespace WinFormsApp2
```

```
{
    public partial class Form1 : Form
    {
        public Form1()
        {
            InitializeComponent();
        }

        private void button1_Click(object sender, EventArgs e)
        {
            string[] names = { "Ana", "Ben", "Carlo", "David" };

            string searchName = txtName.Text.Trim();
            bool found = false;

            foreach (string name in names)
            {
                if (string.Equals(name, searchName,
                    StringComparison.OrdinalIgnoreCase))
                {
                    found = true;
                    break;
                }
            }

            if (found)
            {
                lblResult.Text = "Result: Found!";
            }
            else
            {
                lblResult.Text = "Result: Name not found.";
            }
        }
        private void btnClear_Click(object sender, EventArgs e)
        {
            txtName.Clear();

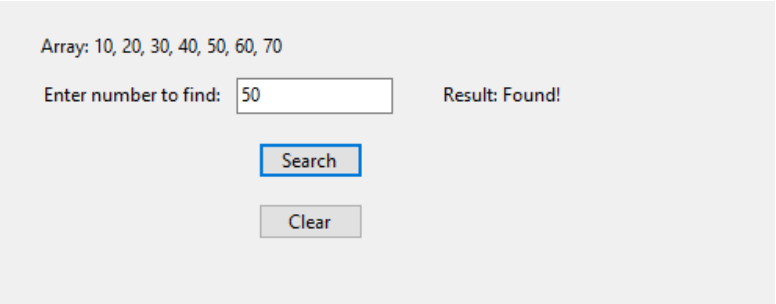
            lblResult.Text = "Result will appear here.";

            txtName.Focus();
        }
    }
}
```

6. 

CODE:

```
7.  namespace WinFormsApp2
8.  {
9.      public partial class Form1 : Form
10.     {
11.         public Form1()
12.         {
13.             InitializeComponent();
14.         }
15.
16.         private void button1_Click(object sender, EventArgs e)
17.         {
18.             int[] numbers = { 5, 10, 15 };
19.             int total = 0;
20.
21.             foreach (int number in numbers)
22.             {
23.                 total = total + number;
24.             }
25.
26.             lblResult.Text = "Total: " + total;
27.
28.         }
29.
30.         private void btnClear_Click(object sender, EventArgs e)
31.         {
32.             {
33.                 lblResult.Text = "Total will appear here.";
34.             }
35.         }
36.     }
37. }
```

7. 

CODE:

namespace WinFormsApp2

```
{
    public partial class Form1 : Form
    {
        public Form1()
        {
            InitializeComponent();
        }

        private void button1_Click(object sender, EventArgs e)
        {
            int[] numbers = { 10, 20, 30, 40, 50, 60, 70 };

            int target;

            if (!int.TryParse(txtSearchNumber.Text, out target))
            {
                lblResults.Text = "Please enter a valid number.";
                return;
            }

            int left = 0;
            int right = numbers.Length - 1;
            bool found = false;

            while (left <= right)
            {
                int middle = (left + right) / 2;

                if (numbers[middle] == target)
                {
                    found = true;
                    break;
                }
                else if (numbers[middle] < target)
                {
                    left = middle + 1;
                }
                else
                {
                    right = middle - 1;
                }
            }

            if (found)
            {
                lblResults.Text = "Result: Found!";
            }
            else
            {
```

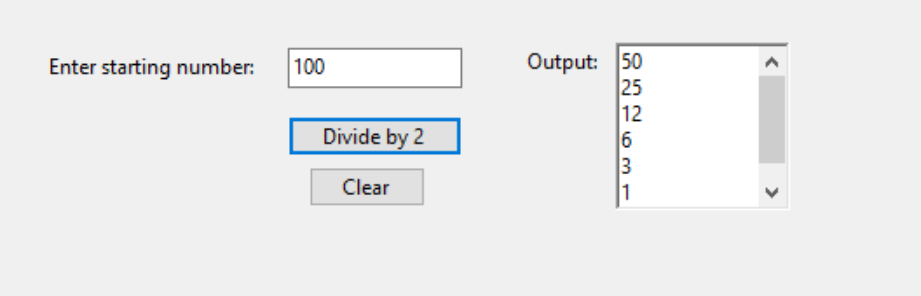
```
            lblResults.Text = "Result: Number not found.";
        }
    }

    private void btnClear_Click(object sender, EventArgs e)
    {
        txtSearchNumber.Clear();

        lblResults.Text = "Result will appear here.";

        txtSearchNumber.Focus();
    }

  }
}
```

8. 

CODE:

```
namespace WinFormsApp2
{
  public partial class Form1 : Form
  {
    public Form1()
    {
      InitializeComponent();
    }

    private void button1_Click(object sender, EventArgs e)
    {
      int n;

      if (!int.TryParse(txtNumber.Text, out n) || n < 1)
      {
        rtbOutput.Clear();
        return;
      }

      rtbOutput.Clear();
```

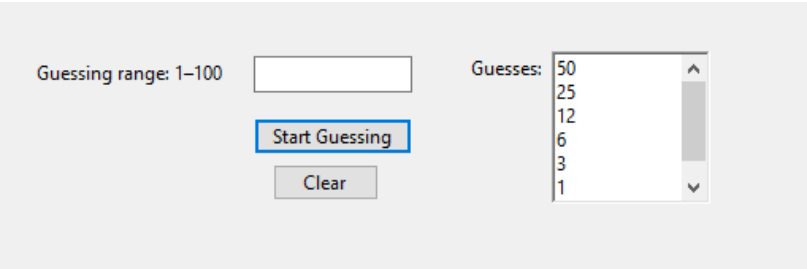
```
            while (n > 1)
            {
                n = n / 2;
                rtbOutput.AppendText(n + Environment.NewLine);

            }
        }
        private void btnClear_Click(object sender, EventArgs e)
        {
            txtNumber.Clear();
            rtbOutput.Clear();

            txtNumber.Focus();
        }

    }

}
```

9. 

CODE:

```
namespace WinFormsApp2
{
    public partial class Form1 : Form
    {
        public Form1()
        {
            InitializeComponent();
        }

        private void button1_Click(object sender, EventArgs e)
        {
            int low = 1;
            int high = 100;

            rtbOutput.Clear();

            while (low <= high)
            {
                int middle = (low + high) / 2;
```

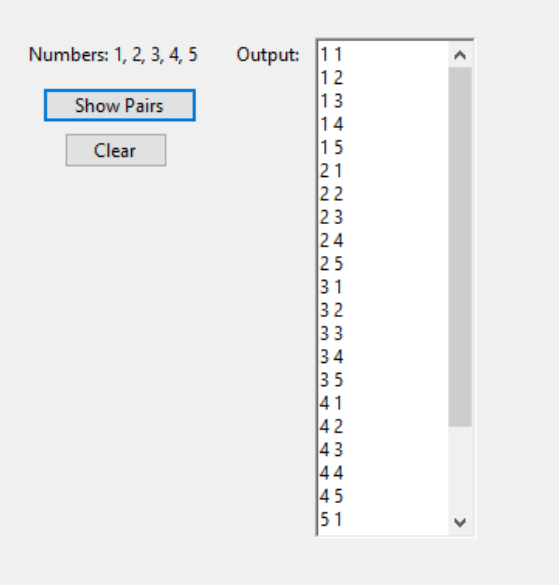
```
            rtbOutput.AppendText(middle + Environment.NewLine);

            high = middle - 1;
        }

    }
    private void btnClear_Click(object sender, EventArgs e)
    {
        rtbOutput.Clear();
    }

  }

}
```

10. 

CODE:

```
namespace WinFormsApp2
{
  public partial class Form1 : Form
  {
    public Form1()
    {
      InitializeComponent();
    }

    private void button1_Click(object sender, EventArgs e)
    {
      int[] numbers = { 1, 2, 3, 4, 5 };
```

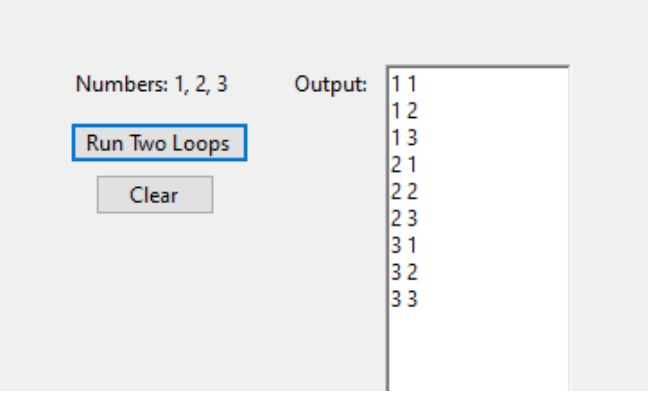
```
            rtbOutput.Clear();

            for (int i = 0; i < numbers.Length; i++)
            {
                for (int j = 0; j < numbers.Length; j++)
                {
                    rtbOutput.AppendText(
                        numbers[i] + " " + numbers[j] +
                        Environment.NewLine);
                }
            }
        }


        private void btnClear_Click(object sender, EventArgs e)
        {
            rtbOutput.Clear();
        }

    }

}
```

11. 

CODE:

```
namespace WinFormsApp2
{
    public partial class Form1 : Form
    {
        public Form1()
        {
            InitializeComponent();
        }

        private void button1_Click(object sender, EventArgs e)
        {
```

```
        int[] numbers = { 1, 2, 3 };

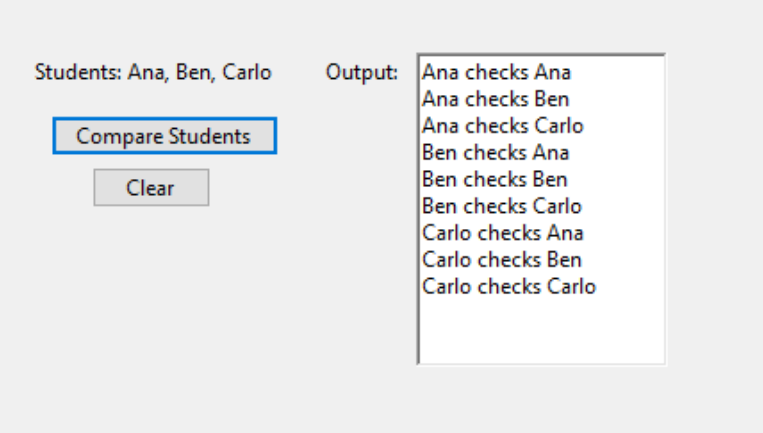
        rtbOutput.Clear();

        for (int i = 0; i < numbers.Length; i++)
        {
            for (int j = 0; j < numbers.Length; j++)
            {
                rtbOutput.AppendText(
                    numbers[i] + " " + numbers[j] +
                    Environment.NewLine);
            }
        }
    }

    private void btnClear_Click(object sender, EventArgs e)
    {
        rtbOutput.Clear();
    }

}

}
```

12. 

CODE:

```
namespace WinFormsApp2
{
    public partial class Form1 : Form
    {
        public Form1()
        {
            InitializeComponent();
        }
```

```
    private void button1_Click(object sender, EventArgs e)
    {
        string[] students = { "Ana", "Ben", "Carlo" };

        rtbOutput.Clear();

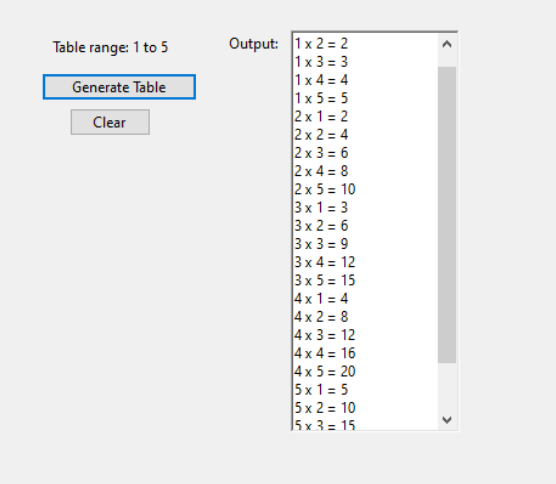
        for (int i = 0; i < students.Length; i++)
        {
            for (int j = 0; j < students.Length; j++)
            {
                rtbOutput.AppendText(
                    students[i] + " checks " + students[j] +
                    Environment.NewLine);
            }
        }
    }


    private void btnClear_Click(object sender, EventArgs e)
    {
        rtbOutput.Clear();
    }

}

}
```

13.



CODE:

```
namespace WinFormsApp2
{
    public partial class Form1 : Form
    {
        public Form1()
```

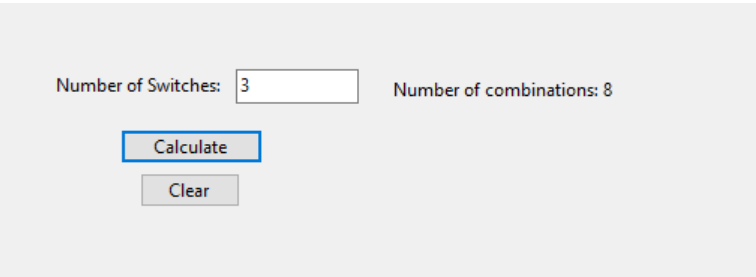
```
    {
        InitializeComponent();
    }

    private void button1_Click(object sender, EventArgs e)
    {
        rtbOutput.Clear();

        for (int i = 1; i <= 5; i++)
        {
            for (int j = 1; j <= 5; j++)
            {
                rtbOutput.AppendText(
                    i + " x " + j + " = " + (i * j) +
                    Environment.NewLine);
            }
        }
    }

    private void btnClear_Click(object sender, EventArgs e)
    {
        rtbOutput.Clear();
    }

}
```

14. 

CODE:

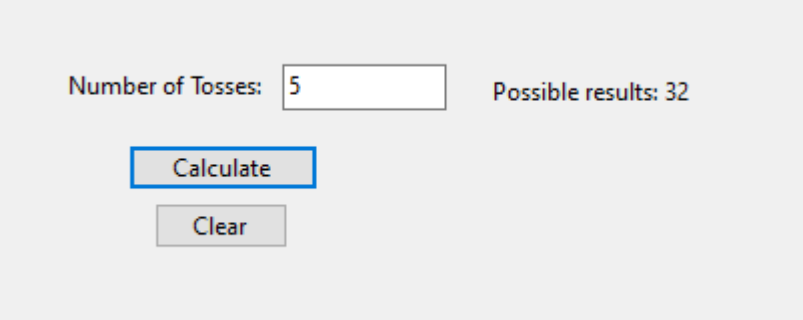
```
namespace WinFormsApp2
{
    public partial class Form1 : Form
    {
        public Form1()
        {
            InitializeComponent();
        }

        private void button1_Click(object sender, EventArgs e)
        {
            if (int.TryParse(txtSwitches.Text, out int n) && n >= 0)
```

```
    {
        int combinations = 1;

        for (int i = 0; i < n; i++)
        {
            combinations *= 2;
        }

        lblResults.Text = "Number of combinations: " + combinations;
    }
    else
    {
        MessageBox.Show("Please enter a valid positive whole number.",
                        "Invalid Input",
                        MessageBoxButtons.OK,
                        MessageBoxIcon.Warning);
    }
}
```

15. 

CODE:

```
namespace WinFormsApp2
{
    public partial class Form1 : Form
    {
        public Form1()
        {
            InitializeComponent();
        }

        private void button1_Click(object sender, EventArgs e)
        {
            if (int.TryParse(txtTosses.Text, out int tosses) && tosses >= 0)
            {
                int possibilities = 1;

                for (int i = 0; i < tosses; i++)
                {
```

```
            possibilities *= 2;
        }

        lblResults.Text = "Possible results: " + possibilities;

    }
}
```